

Homework 2

MATH 346/377 - Winter 2025

Due on Sunday, February 9th, 9:00pm on Crowdmark

Math 346: Do problems 1 to 7 – Total of points= 85pts

Math 377: Do problems 1 to 9 – Total of points= 110pts

1. (8 points) We say that a positive integer n is a perfect k -th power if $n = m^k$ for some integer m .
 - (a) (4 points) Suppose that a, b are coprime positive integers. Show that the product ab is a perfect k -th power if and only if a and b are both perfect k -th powers.
 - (b) (4 points) Is the statement still correct if a and b are not positive? or if a and b are not coprime? Prove or give a counterexample.

2. (15 points) Are the following statements true or false, where a and b are positive integers and p is prime? In each case, give a proof or a counterexample.
 - (a) (3 points) If $\gcd(a, p^2) = p$ then $\gcd(a^2, p^2) = p^2$.
 - (b) (3 points) If $\gcd(a, p^2) = p$ and $\gcd(b, p^2) = p^2$ then $\gcd(ab, p^4) = p^3$.
 - (c) (3 points) If $\gcd(a, p^2) = p$ and $\gcd(b, p^2) = p$ then $\gcd(ab, p^4) = p^2$.
 - (d) (3 points) If $\gcd(a, p^2) = p$ then $\gcd(a + p, p^2) = p$.
 - (e) (3 points) We have $\gcd(a^n, b^n) = \gcd(a, b)^n$.

We have seen that there are infinitely many primes. Since 2 is the only even prime, it is immediate that there are infinitely many odd primes i.e. there are infinitely many primes $p \equiv 1 \pmod{2}$. This has been vastly generalized by Dirichlet:

Theorem (Dirichlet's theorem on primes in arithmetic progressions). *Let a and b be two positive coprime integers. There are infinitely many primes $p \equiv a \pmod{b}$ i.e. of the form $p = aq + b$.*

Although it is a difficult theorem to prove in general, the following two exercises prove some special cases.

3. (10 points) Follow Euclid's proof¹ to show that there are infinitely many primes $p \equiv 2 \pmod{3}$.

Numbers of the form $F_n := 2^{2^n} + 1$ are called Fermat numbers. The only primes of the form $2^m + 1$ must be Fermat numbers, by part a) below. Fermat conjectured that F_n is prime for every $n \geq 0$. This is true for the first few Fermat numbers $F_0 = 3$, $F_1 = 5$, $F_2 = 17$, $F_3 = 257$ and $F_4 = 65537$ which are primes. But in 1732, Euler showed that the next Fermat number

$$F_5 = 2^{2^5} + 1 = 4294967297 = 641 \times 6700417 \quad (5)$$

is not prime. (Observe that the two primes 641 and 6700417 are both congruent to 1 mod 64; see part b) below.) No other prime Fermat number is known.

4. (17 points) The goal of this problem is to use Fermat numbers to show that there are infinitely many primes that are congruent to 1 mod 2^k for any $k \geq 1$.
- (a) (5 points) Show that if $2^m + 1$ is prime, then m is a power of 2.
- (b) (5 points) Show that if a prime p divides the Fermat number F_n , then $p \equiv 1 \pmod{2^{n+1}}$.

Hint: Fermat's little theorem.

- (c) (7 points) Show that distinct Fermat numbers are coprime. For a fixed $k \geq 1$, deduce from b) and c) that there are infinitely many primes that are congruent to 1 mod 2^k .

Hint: First, find a recursion formula for the Fermat numbers, and then a formula for the product $F_0 F_1 \cdots F_n$.

5. (12 points) Let $a, b \geq 2$ be coprime integers, and $n \geq 3$ an even integer. Show that $a + b$ divides $a^{n+1} + b^{n+1}$ and that

$$\gcd\left(a + b, \frac{a^{n+1} + b^{n+1}}{a + b}\right) \text{ divides } n + 1. \quad (16)$$

Hint: Consider the polynomial division of $\frac{x^{n+1} + b^{n+1}}{x + b}$ by $x + b$.

¹This is the proof we have seen in class: assume there are only finitely many primes $p_1, \dots, p_r$. The integer $p_1 \cdots p_r + 1$ is divisible by a prime that is not in the list, and this is a contradiction.

6. (11 points) Let p be a prime. Show that $\sqrt{2} + \sqrt{p+2}$ is always an irrational number.

Let $\pi(x)$ be the number of primes $p \leq x$. So for example $\pi(1) = 0$, $\pi(2) = 1$, $\pi(10) = 4$ and $\pi(20) = 8$ (the primes less than 20 are 2, 3, 5, 7, 11, 13, 17, 19). In class I have mentioned the Prime Number Theorem, that states that

$$\lim_{n \rightarrow \infty} \frac{\pi(n) \log(n)}{n} = 1. \quad (33)$$

It is known that there are always at least a prime between n and $2n$, and the PNT implies that the density goes to 0 as $n \rightarrow \infty$. More precisely, we have

$$\lim_{n \rightarrow \infty} \frac{|\{\text{primes } p, n < p \leq 2n\}|}{n} = \lim_{n \rightarrow \infty} \frac{\pi(2n) - \pi(n)}{n} = 0. \quad (34)$$

The following exercise proves this fact without using the PNT.

7. (12 points) Show that if p is a prime with $n < p \leq 2n$ then p divides the binomial coefficient $\binom{2n}{n}$. Deduce that

$$n^{\pi(2n) - \pi(n)} < 2^{2n}, \quad (35)$$

and hence $\frac{\pi(2n) - \pi(n)}{n} < 2 \log(2) / \log(n)$. (This shows that the limit goes to 0 as $n \rightarrow \infty$, hence the density of primes between n and $2n$ goes to 0).

Problems 8 and 9 for MATH 377 students.

We have seen in class that “there are more primes than squares”, which can be deduced from the PNT. This can also be seen from the divergence of

$$\lim_{N \rightarrow \infty} \sum_{i=1}^N \frac{1}{p_i} = \frac{1}{2} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} \cdots = \infty \quad (40)$$

where p_i is the i -th prime number, since on the other hand the sum $\sum_{k=1}^{\infty} 1/k^2$ converges to $\pi/6$ (You can assume this fact without proof. Note also that $\sum_{k=1}^{\infty} 1/k$ diverges). The following exercise is a proof of the divergence of (40).

8. (12 points) An integer $n > 1$ is said to be square-free² if any prime appearing in its factorization appears with exponent 1. Show that any integer $n > 1$ can be written

²The integer $105 = 3 \cdot 5 \cdot 7$ is square-free, but $54 = 2 \cdot 3^3$ is not.

$n = k^2m$ where m is square-free, and that we have

$$\lim_{N \rightarrow \infty} \sum_{\substack{1 \leq m \leq N \\ m \text{ squarefree}}} \frac{1}{m} = \infty. \quad (41)$$

Deduce that

$$\lim_{N \rightarrow \infty} \prod_{i=1}^N \left(1 + \frac{1}{p_i}\right) = \infty, \quad (42)$$

and conclude the divergence of (40).

Hint: We have $1 + x \leq e^x$.

We have seen that we can use factorization in ring of integers to solve diophantine equations, for example the equation $2x^3 = y^2 + 1$. It relied on the following proposition, which is an immediate generalization of Exercise 2. (The proof is done in the exact same way by using the unique factorization into irreducibles, but you can use it without proof in the next problem.)

Proposition. *Let R be a Euclidean domain and suppose that we have $\alpha_1\alpha_2 = \gamma^k$ for some nonzero $\alpha_1, \alpha_2, \gamma \in R$. If α_1 and α_2 are coprime, then there exist elements $\gamma_1, \gamma_2 \in R$ and units³ $u_1, u_2 \in R^\times$ such that $\alpha_i = u_i\gamma_i^k$.*

9. (13 points) The ring $\mathbb{Z}[\sqrt{-2}]$ is a Euclidean domain with Euclidean function $|a + \sqrt{-2}ib|^2 = a^2 + 2b^2$. (You can assume it without proof). We want to find all integers $x, y \in \mathbb{Z}$ satisfying the equation $x^3 = y^2 + 2$.
 - (a) (2 points) Show that the group of units is $\mathbb{Z}[\sqrt{-2}]^\times = \{\pm 1\}$ and that $i\sqrt{2}$ is irreducible.
 - (b) (4 points) Reduce the equation modulo 4 to deduce that x and y are both odd.
 - (c) (7 points) Following the example that we have seen in class solve the equation $x^3 = y^2 + 2$.
Hints: You should find two solutions. You can use the above proposition without proof.

³In class I only considered the case where the unit is 1 because in $\mathbb{Z}[i]$ any unit is a cube ($1^3 = 1, (-1)^3 = -1, i^3 = -i$ and $(-i)^3 = i$). The same happens in the exercise where you can also ignore the unit)